

Laboratory Audit of (lab name and lab identifier):

ISO/IEC 17025 Cert. identifier: [\[see OCM opening / closing conference form\]](#)

Audit/Inspection Date(s): [\[see OCM opening / closing conference form\]](#)

OCM Auditor(s)/Inspectors: [\[see OCM opening / closing conference form\]](#)

This checklist is based on the OCM Cannabis Laboratory Quality System Standard (LQSS).

MICROBIOLOGY TESTING STANDARDS

ID	Citation	Does the laboratory comply with this section?	Yes	No	N/A	Notes
REAGENTS, REFERENCE STANDARDS, AND MEDIA						
1	LQSS, Section IV	Maintain a consumables log or inventory for all reagents, reference standards and media purchased and received.				
2	LQSS, Section IV	All reagents and reference standards, including any working standards: 1. labeled to indicate identity, batch number, date received or prepared, expiration date, and where applicable, concentration or purity, and date opened; 2. stored under appropriate conditions to minimize degradation or deterioration of the material; 3. within their expiration or re-qualification dates at the time of use; and 4. documented on records for each analysis.				
3	LQSS, Section IV	Deteriorated or outdated reagents, reference standards and media must be properly discarded.				
4	LQSS, Section IV	All reference standards must be acquired from certified sources.				
5	LQSS, Section IV	Must obtain, or for internally produced standards, create a certificate of conformance for each lot of reference standard, and be kept on file.				
6	LQSS, Section IV, B	For a microbiological laboratory, the reagent water quality and media checks that must be done are listed below. 1) Monitor quality as per Standard Method Table 9020:II. 2) Keep certificate of analysis from laboratory testing the water quality.				

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REAGENTS, REFERENCE STANDARDS, AND MEDIA																																							
7	LQSS, Section IV, B	TABLE 9020.II. QUALITY OF REAGENT WATER USED IN MICROBIOLOGY TESTING <table border="1"> <thead> <tr> <th>Test</th><th>Monitoring Frequency</th><th>Maximum Acceptable Limit</th></tr> </thead> <tbody> <tr> <td>Chemical tests:</td><td></td><td></td></tr> <tr> <td>Conductivity</td><td>Continuously or usage day</td><td><2 µmhos/cm (µmhos/cm) at 25°C</td></tr> <tr> <td>Total organic carbon</td><td>Monthly</td><td><1.0 mg/L</td></tr> <tr> <td>Heavy metals, single (Cd, Cr, Cu, Ni, Pb, and Zn)</td><td>Annually*</td><td><0.05 mg/L</td></tr> <tr> <td>Heavy metals, total</td><td>Annually*</td><td><0.10 mg/L</td></tr> <tr> <td>Total chlorine residual</td><td>Monthly or with each use</td><td><0.1 mg/L</td></tr> <tr> <td>Bacteriological tests:</td><td></td><td></td></tr> <tr> <td>Heterotrophic plate count†</td><td>Monthly</td><td><500 CFU/mL or MPN <500/mL</td></tr> <tr> <td>Use test [see 9020B.5(2)]</td><td>For a new source</td><td>Student's <i>t</i> ≤ 2.78</td></tr> <tr> <td>Water quality test [see 9020B.5(1)]‡</td><td>Annually</td><td>0.8–3.0 ratio</td></tr> </tbody> </table> * Or more frequently if there is a problem. † See Section 9215. ‡ This water-quality test is not needed for Type II water or better, as defined in <i>Standard Methods</i> (18th and 19th Editions), Section 1080C, or medium quality water or better, as defined in <i>Standard Methods</i> (20th, 21st, 22nd, and Online Editions), Section 1080C.	Test	Monitoring Frequency	Maximum Acceptable Limit	Chemical tests:			Conductivity	Continuously or usage day	<2 µmhos/cm (µmhos/cm) at 25°C	Total organic carbon	Monthly	<1.0 mg/L	Heavy metals, single (Cd, Cr, Cu, Ni, Pb, and Zn)	Annually*	<0.05 mg/L	Heavy metals, total	Annually*	<0.10 mg/L	Total chlorine residual	Monthly or with each use	<0.1 mg/L	Bacteriological tests:			Heterotrophic plate count†	Monthly	<500 CFU/mL or MPN <500/mL	Use test [see 9020B.5(2)]	For a new source	Student's <i>t</i> ≤ 2.78	Water quality test [see 9020B.5(1)]‡	Annually	0.8–3.0 ratio				
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MICROBIOLOGY-SPECIFIC QC																																							
Commercially available; Ready to Use Reagent, Reference Standard and Media																																							
8	LQSS, Section IV, Table 1	Maintain certificate of conformance (COC) for each new lot to document the following:																																					
9	LQSS, Section IV, Table 1	- Cell/CFU concentration of Certified Reference Materials																																					
10	LQSS, Section IV, Table 1	- Sterility, pH and volume accuracy of Dilution Buffers																																					
11	LQSS, Section IV, Table 1	- Sterility, growth promotion, growth inhibition, appearance, and pH (or per manufacturer's instructions for pH) of Media Performance																																					
12	LQSS, Section IV, Table 1	Document with each new lot , the following: 1.) preparation of new working stock from stock culture, 2.) media appearance , and 3.) count comparisons of old and new lots.																																					
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Automated MPN Instrumentation						
16	LQSS, Section IV, Table 1	Measure performance monthly or per manufacturer's instructions.				
MICROBIOLOGY TESTING STANDARDS						
Culture Methods and MPN Instrumentation						
Qualitative Culturing						
17	LQSS Section VI, C. 1) Table 4	Positive Control: Analyze a Matrix Spike (MS) per batch or on a regular basis (i.e., monthly based on historical data).				
18	LQSS, Section IV	Lot number of the reference standard used is recorded in the documentation for each analysis.				
19	LQSS Section VI, C. 1)	The control meets the acceptance criteria of either the CRM growth in CFU/g or growth of the target organism in the matrix spike.				
20	LQSS Section VI, C. 1) Table 4	Negative Control: Analyze a Method Blank (MB) per batch.				
21	LQSS, Section IV	Lot number of the reference standard used is recorded in the documentation for each analysis.				
22	LQSS Section VI, C. 1)	The control is negative for all organisms and growth.				
Quantitative Culturing						
22	LQSS Section VI, C. 1) Table 4	Positive Control: Analyze a <u>MS (true organism)</u> for TAMC and TYAM per batch or on a regular basis (i.e., monthly based on historical data).				
23	LQSS, Section IV	Lot number of the reference standard used is recorded in the documentation for each analysis.				
24	LQSS Section VI, C. 1)	The control meets the acceptance criteria of either the CRM growth in CFU/g or growth of the target organism in the matrix spike.				
25	LQSS Section VI, C. 1) Table 4	Positive Control: Alternatively, analyze Sample Duplicates (for samples with expected values > 25 cfu/g) per batch or on a regular basis (i.e., monthly based on historical data).				
26	LQSS, Section IV	Lot number of the reference standard used is recorded in the documentation for each analysis.				

ID	Citation	Does the laboratory comply with this section?	Yes	No	N/A	Notes
Culture Methods and MPN Instrumentation						
26	LQSS Section VI, C. 1)	The control meets the acceptance criteria of either the CRM growth in CFU/g or growth of the target organism in the matrix spike.				
27	LQSS Section VI, C. 1) Table 4	Positive Control: Analyze a <u>CRM (in actual matrix)</u> per batch.				
28	LQSS, Section IV	Lot number of the reference standard used is recorded in the documentation for each analysis.				
29	LQSS Section VI, C. 1)	The control meets the acceptance criteria of either the CRM growth in CFU/g or growth of the target organism in the matrix spike.				
30	LQSS Section VI, C. 1) Table 4	Method Blank (MB)/Negative Control: Incubate and analyze a dilution buffer or equivalent for TAMC and TYAM <u>per batch</u> .				
31	LQSS, Section IV	Lot number of the reference standard used is recorded in the documentation for each analysis.				
32	LQSS Section VI, C. 1)	The control is negative for all organisms and growth.				
33	LQSS Section VI, C. 1) Table 4	Analyst Quantitative Performance: Document plate count comparisons <u>monthly</u> (2 or > w/in 10%; 1 w/in 5%)				
34	LQSS Section VI, C. 1)	For each matrix or form, the minimum amount of historical data is 50 matrix spikes. Evaluate the historical data against the acceptance limits for the certified reference material. The data must show the matrix does not contribute to suppression / enhancement. The matrix or forms of cannabis include, but are not limited to flower, oils for vaporization, chews, gummies, beverages, and chocolates.				

ID	Citation	Does the laboratory comply with this section?	Yes	No	N/A	Notes
Molecular Assays/Methods (i.e., polymerase chain reaction (PCR))						
34	LQSS Section VI, C. 2), Table 5	Analyze PCR Positive Control per each PCR assay run				
35	LQSS, Section IV	Lot number of the reference standard used is recorded in the documentation for each analysis.				
36	LQSS Section VI, C. 2), Table 5	Analyze a PCR Inhibition Positive Control (Internal Sample Control) in every extracted sample				
37	LQSS Section VI, C. 2), Table 5	Analyze a MS with each batch or on a regular basis (i.e., monthly based on historical data).				
38	LQSS, Section IV	Lot number of the reference standard used is recorded in the documentation for each analysis.				
39	LQSS Section VI, C. 2), Table 5	Analyze PCR Negative Control with each PCR assay run: Non-target DNA check for non-target amplification				
40	LQSS, Section IV	Lot number of the reference standard used is recorded in the documentation for each analysis.				
41	LQSS Section VI, C. 2), Table 5	Analyze a No Template Control with each PCR assay run: For determination of master mix contamination				
42	LQSS Section VI, C. 2), Table 5	Analyze a MB once per extraction batch: For determination of process and analysis contamination using sterile water				
43	LQSS, Section IV	Lot number of the reference standard used is recorded in the documentation for each analysis.				
44	LQSS Section VI, C. 2)	For each matrix or form, the minimum amount of historical data is 50 matrix spikes. Evaluate the historical data against the acceptance limits for the certified reference material. The data must show the matrix does not contribute to suppression / enhancement. The matrix or forms of cannabis include, but are not limited to flower, oils for vaporization, chews, gummies, beverages, and chocolates.				

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QC for MALTI-TOF						
45	LQSS Section VI, C. 3)	Perform automatic instrument calibration using a manufacturer-specified calibration standard before every run, each day of testing, when a new target is used, or more often if the manufacturer recommends it				
46	LQSS Section VI, C. 3)	Documents calibration results and promptly investigate calibration failures				
47	LQSS Section VI, C. 3)	Analyzes positive external control (an appropriate control micro-organism or calibrator) with each extraction technique				
48	LQSS Section VI, C. 3)	For yeast/molds, appropriate QC organisms for each organism type must be run each day they test for these microorganisms				
49	LQSS Section VI, C. 3)	Analyzes negative control (method blank) with each extraction technique consisting of water without cells spotted directly on the target plate or slide				
50	LQSS Section VI, C. 3)	Operates a platform with disposable or reusable targets and test a blank or negative control to ensure adequate cleaning of the target and before sample analyses				
51	LQSS Section VI, C. 3)	If an extraction is performed, the reagents used for the extraction are spotted and overlaid with matrix to ensure no false-positive results are produced due to reagent contamination				
52	LQSS Section VI, C. 3)	Test lysis buffers and other reagents used for sample preparation to ensure they are free of contamination per each lot of buffer and other reagents used				
53	LQSS Section VI, C. 3)	Follow manufacturer's recommendations for optimal culture conditions, sample preparation and media types				
54	LQSS Section VI, C. 3)	Uses isolates that are grown and maintained according to the manufacturer's instructions				
55	LQSS Section VI, C. 3)	Has extensively validated lower confidence thresholds, and/or adheres to the manufacturer-recommended thresholds for genus and species-level identifications				
Probe-based Quantitative PCR and Melting Curve Analysis						
56	LQSS Section VI, C. 2)	A sequencing reaction control using a template and primer provided by the manufacturer is run according to the manufacturers' instruction In the event of unacceptable results from sequence analyse				